

In order to model this scenario in Python, we can define a function. We'll start with a hot temperature and a willingness to explore and go downward, and we'll end with a cool temperature and an unwillingness to go downward. Our temperature function is relatively simple. It takes t as an argument, where t stands for time:

```
temperature = lambda t: 1/(t + 1)
```

You can see a simple plot of the temperature function by running the following code in the Python console. This code starts by importing `matplotlib` functionality and then defines `ts`, a variable containing a range of t values between 1 and 100. Finally, it plots the temperature associated with each t value. Again, we're not worried about units or exact magnitude here because this is a hypothetical situation meant to show the general shape of a cooling function. So we use 1 to represent our maximum temperature, 0 to represent our minimum temperature, 0 to represent our minimum time, and 99 to represent our maximum time, without specifying units.

```
import matplotlib.pyplot as plt
ts = list(range(0,100))
plt.plot(ts, [temperature(t) for t in ts])
plt.title('The Temperature Function')
plt.xlabel('Time')
plt.ylabel('Temperature')
plt.show()
```

The plot looks like [Figure 6-4](#).